

Unsupervised Domain Adaptation for Adverse Weather Conditions

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Type of work: Groups Studies

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Motivation

Autonomous driving systems and intelligent transportation technologies require robust visual perception in all environmental conditions. A core component of such perception is semantic segmentation, where each pixel in a scene is classified into categories such as road, vehicle, pedestrian, building, or sky. Accurate segmentation is essential for downstream tasks including path planning, obstacle avoidance, traffic interaction, and safe navigation.

However, most state-of-the-art semantic segmentation models are trained on labeled clear-weather datasets such as Cityscapes and assume favorable daytime visibility. When deployed in real-world adverse conditions such as fog, rain, snow, or nighttime scenes, their performance often degrades significantly. Reduced visibility, illumination changes, reflections, motion blur, and weather artifacts create a strong domain shift between training and deployment environments. Furthermore, dense pixel-level annotation for adverse-weather scenes is costly, time-consuming, and difficult to scale.

Solving this problem is important for both scientific and practical reasons. From a societal perspective, robust perception under adverse weather can improve the safety and reliability of autonomous vehicles and driver assistance systems. From an industrial perspective, it supports real-world deployment readiness for automotive and robotics applications. From a research perspective, it advances methods for learning under limited supervision and challenging distribution shifts.

This topic is especially timely due to the rapid progress of transformer-based vision architectures, self-training strategies, and domain adaptation methods such as DAFormer and MIC. At the same time, increasing interest in trustworthy AI and robust autonomous systems makes adaptation to real-world weather conditions an urgent and relevant challenge.

Recent domain adaptation methods such as DAFormer and MIC have shown strong performance [1, 2].

Task Description

The goal of this research project is to investigate how semantic segmentation models trained on labeled clear-weather data can be effectively adapted to adverse-weather environments where no pixel-level annotations are available. In particular, the project focuses on transferring knowledge from the Cityscapes dataset as the labeled source domain to the ACDC dataset as the unlabeled target domain containing fog, rain, snow, and night scenes.

The project aims to implement, reproduce, and compare state-of-the-art unsupervised domain adaptation methods such as DAFormer and MIC. These approaches use self-training, pseudo-label generation, transformer-based feature learning, and consistency regularization to reduce the domain gap between clear-weather and adverse-weather scenes.

A further objective is to investigate whether weather-aware masking strategies can improve masked consistency learning. Instead of using only generic random masks, the project will explore masking patterns that simulate realistic weather degradations, such as haze for fog, streak-like patterns for rain, dark-region masking for night scenes, and white occlusions for snow.

The main research questions of this project are:

- How severe is the performance degradation when clear-weather segmentation models are applied directly to adverse-weather domains?
- How effectively do DAFormer and MIC improve segmentation performance on fog, rain, snow, and night scenes?
- Can weather-aware masking strategies further improve robustness compared to generic masking approaches?
- Which weather conditions remain the most challenging after adaptation?
- Is it possible to develop one robust model that performs consistently across multiple adverse-weather domains?

The expected outcomes of the project include implemented and trained baseline models, comparative benchmark results, qualitative segmentation visualizations, and an improved adaptation strategy for robust scene understanding under adverse weather conditions.

Scientific Approach

The project will follow an experimental research methodology based on implementation, reproduction of existing methods, and evaluation on benchmark datasets.

- Which sources will be used?

The primary datasets used in this project will be the Cityscapes dataset and the ACDC dataset. Cityscapes will serve as the labeled source domain containing clear-weather urban street scenes with pixel-level annotations. ACDC will be used as the target domain containing adverse-weather scenes such as fog, rain, snow, and night conditions.

The scientific literature provided by the supervisor includes important baseline methods such as DAFormer and MIC, as well as recent work on semantic segmentation under multiple adverse weather conditions. Additional related publications on unsupervised domain adaptation, self-training, pseudo-labeling, and transformer-based segmentation models will also be reviewed.

These sources will be used to understand current state-of-the-art approaches, reproduce benchmark baselines, and identify research gaps for further improvement.

- **Which methods are planned?**

The project will begin with a baseline semantic segmentation model trained only on Cityscapes. This model will be evaluated directly on ACDC to quantify the domain shift.

Afterwards, the following unsupervised domain adaptation methods will be implemented and compared:

- DAFormer: Transformer-based domain adaptation using pseudo-label self-training.
- MIC: Masked Image Consistency for robust context-aware adaptation.

In addition, a project-specific extension will be investigated based on weather-aware masking strategies. These masking patterns will simulate realistic degradations caused by fog, rain, snow, and night scenes.

- **What experiments and implementation are planned?**

The implementation will be carried out in Python using PyTorch and MMSegmentation. Training and experiments will be performed on available GPU resources such as a local workstation or university HPC cluster.

Planned experiments include:

- Training a clear-weather baseline model on Cityscapes.
- Evaluating baseline performance on ACDC.
- Reproducing DAFormer and MIC results.
- Comparing methods across fog, rain, snow, and night subsets.
- Testing the proposed weather-aware masking extension.
- Conducting ablation studies on masking strategies and confidence thresholds.

- **Which metrics will be used?**

The main evaluation metric will be Mean Intersection over Union (mIoU), which is the standard metric for semantic segmentation.

Additional metrics include:

- Per-class IoU
- Pixel accuracy
- Performance separated by weather condition
- Qualitative visual comparison of segmentation outputs

These metrics will allow both quantitative and qualitative assessment of adaptation performance.

Time Schedule

Estimate how you want to use your time until submission.

Time period	Tasks
Week 1	Finalizing and submitting this exposé, project planning, literature survey, and resource setup.
Weeks 2–4	Detailed literature review on semantic segmentation, domain adaptation, DAFormer, MIC, and adverse-weather perception. Downloading and preparing Cityscapes and ACDC datasets.
Weeks 5–7	Implementation of baseline semantic segmentation model trained on Cityscapes. Initial evaluation on ACDC to analyze domain shift.
Weeks 8–11	Reproduction and training of DAFormer. Hyperparameter tuning and baseline comparison.
Weeks 12–15	Reproduction and evaluation of MIC. Comparative analysis across fog, rain, snow, and night subsets.
Weeks 16–19	Development and implementation of the proposed weather-aware masking extension. Ablation studies and refinement.
Weeks 20–22	Full experimental evaluation, result analysis, generation of tables, figures, and qualitative visualizations.
Weeks 23–25	Writing of final report, methodology chapter, experiments chapter, proofreading, and revisions.
Week 26	Final corrections, submission of final project report, and preparation of presentation if required.

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